JSS MAHAVIDYAPEETHA

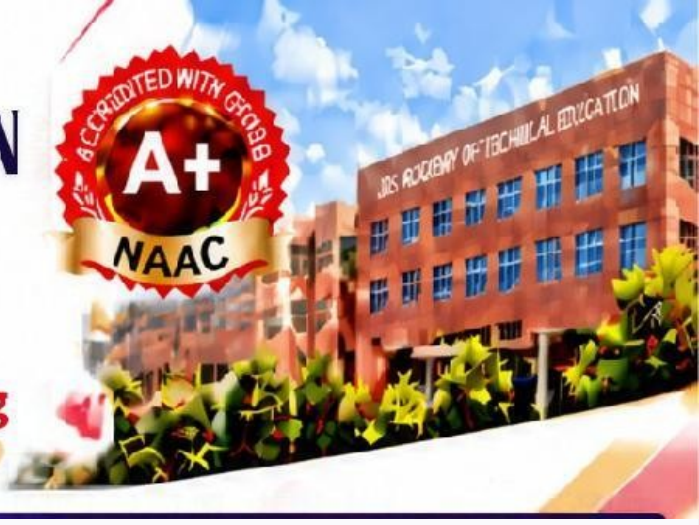
JSS ACADEMY OF TECHNICAL EDUCATION

BENGALURU

Open Day 2026

Department of Computer Science & Engineering

FINAL YEAR PROJECT EXHIBITION



TEAM MEMBERS & PROJECT GUIDE

STUDENTS (GROUP 43)

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PROJECT GUIDE DETAILS

Dr. P. B. Mallikarjuna
Professor
Department of Computer Science & Engineering
JSS Academy of Technical Education
Bengaluru – 560060, Karnataka
Lab Venue: A 204 | Academic Block A

PROJECT TITLE & IDENTITY

Explainable Reinforcement Learning-Based Intrusion Detection and Autonomous Response System (XRL-IDARS)

Autonomous Network Security • Explainable AI • Deep Q-Learning

- **Project Group: 43 | Lab: A 204 | Academic Year: 2025–2026**
- Thematic Area: Cybersecurity, Privacy & Digital Trust
- Operational Core: Random Forest (Sole Live ML Detector) + TreeSHAP (<2.5 ms) + Dueling DQN [4 Actions] + Deterministic Safety Gate
- **Live Interface: Real-Time Streamlit SOC Dashboard (Port :8501)**

ABSTRACT

Enterprise Security Operations Centers (SOCs) face crippling alert fatigue from tens of thousands of daily intrusion warnings, while static IPS policies lack adaptive containment and induce severe collateral service outages.
XRL-IDARS resolves this challenge through a decoupled architecture: sub-millisecond Random Forest classification on 10 deterministic flow statistics (R10), instantaneous local TreeSHAP attribution (<2.5 ms), and a causal 6-dimensional Dueling Deep Q-Network (DQN) response policy. Operating under an inviolable deterministic safety gate, the system yields +64.3% cost reduction over baseline and 100% critical-host invariant compliance without kernel tampering.

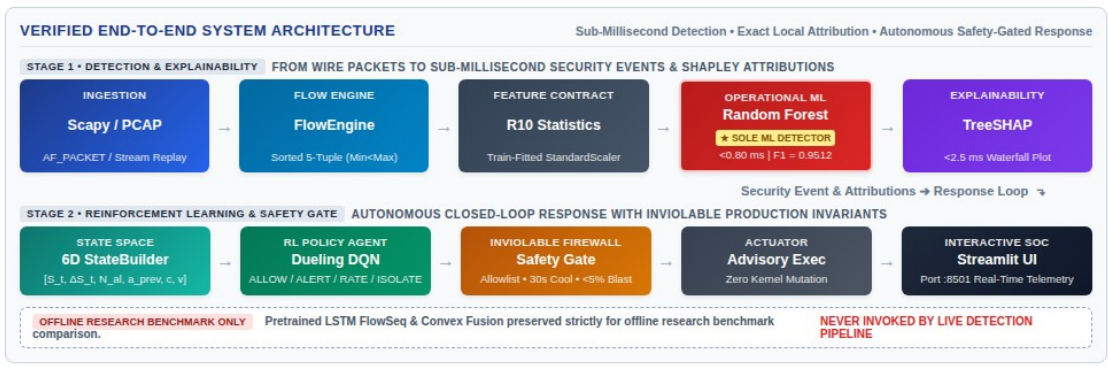
PROBLEM STATEMENT

- **Alert Fatigue & Burst Drops:** Deep temporal models (LSTMs >12 ms) suffer severe packet drops under line-rate traffic bursts.
- **Opaque Black-Box Detection:** Standard ML detectors fail to supply immediate, human-verifiable Shapley feature attributions during incidents.
- **Destructive Collateral Outages:** Binary IPS drop rules isolate mission-critical servers, producing severe unintended service downtime.
- **Non-Causal RL State Leakage:** Prior RL intrusion response architectures suffer temporal data leakage and exhibit severe policy chattering.

OBJECTIVES

1. **Sub-Millisecond ML Detection:** Train operational RF on R10 flow statistics (<0.80 ms latency, F1 >= 0.95).
2. **Real-Time Explainability:** Deliver exact TreeSHAP local attributions in <2.5 ms for top contributing features.
3. **Multi-Tier RL Response Policy:** Formulate causal 6D MDP across 4 discrete actions (ALLOW, ALERT, RATE_LIMIT, ISOLATE).
4. **Inviolable Safety Enforcement:** Enforce 100% immunity on critical assets via deterministic allowlists & blast circuit breaker.
5. **Interactive SOC Telemetry:** Deploy real-time Streamlit SOC dashboard (:8501) with Live, PCAP, and Benchmark modes.

SYSTEM ARCHITECTURE / BLOCK DIAGRAM



METHODOLOGY

1. **Packet Capture & PCAP Stream Replay** `SCAPY / AF_PACKET`
   Continuous wire packet capture and lossless PCAP replay; non-blocking buffer queuing.
2. **Canonical Flow Demarcation & R10 Extraction** `FLOWENGINE`
   Bidirectional flow aggregation using sorted 5-tuple (A<B); 10-feature statistical contract with train-fitted StandardScaler.
3. **Sub-Millisecond RF Detection & TreeSHAP** `OPERATIONAL ML`
   100-tree Random Forest classifies flows in <0.80 ms (F1=0.9512); TreeSHAP computes exact local attributions in <2.5 ms.
4. **Causal 6D StateBuilder & Dueling DQN Policy** `DEEP RL`
   Builds strictly causal state s_t=[S_t, ΔS_t, N_al, a_prev, c, v]; Dueling DQN selects optimal action from 4 discrete tiers.
5. **Deterministic Safety Gate & Advisory Actuation** `ZERO-MUTATION`
   Enforces critical allowlists, 30s cooldown, and 5% blast-radius circuit breaker; logs advisory recommendations to Streamlit SOC.

RESULTS AND PERFORMANCE ANALYSIS

1. OPERATIONAL RANDOM FOREST DETECTION (HELD-OUT TEST SETS)

EVALUATION BENCHMARK	FLOW COUNT (N)	ACCURACY	F1-SCORE	ROC-AUC	INFERENCE LATENCY
CSE-CIC-IDS2017 (D_pol_test, R10)	355,865	99.06%	**0.9512**	0.9989	**< 0.80 ms (780 µs)**
CSE-CIC-IDS2018 (Multi-Class, R10)	1,662,419	97.43%	**0.8861**	0.9966	**< 0.85 ms**

+64.30%	+28.31%	100%	0.20%
COST REDUCTION VS ALWAYS-ALLOW (P < 0.001)	COST REDUCTION VS SINGLE-THRESHOLD (P < 0.001)	CRITICAL HOST INVARIANT ADHERENCE (375/375)	FALSE QUARANTINE RATE (LIMIT < 2.0%)

2. PRE-REGISTERED CRITERIA & HONEST SCIENTIFIC DISCLOSURE (315 RUNS, 90 BOOTSTRAP TESTS)

RESEARCH QUESTION & PRE-REGISTERED CRITERION	TARGET LIMIT	OBSERVED EMPIRICAL RESULT	SCIENTIFIC FINDING
RQ7: Cost Reduction vs Best Baseline	≥ 15.0% (p < 0.01)	+28.31% vs B1 (p<0.001), +0.18% vs B2 (p=0.7600)	H1 SUPPORTED (p<0.001) · <15% TARGET NOT MET
RQ7.1: Action Chattering Index (ACI)	ACI < 0.01 (≤1 jump/100)	ACI = 0.0214 (4-Action) / ACI = 0.0019 (3-Action)	FAIL (4-Action) · PASS (3-Action)
RQ7.2: False Quarantine Rate (FQR)	FQR < 2.0% on benign	FQR = 1.00% – 1.58% (0.20% test population)	PASS (ZERO OUTAGES)
RQ7.3: Safety Gate Compliance	100% invariant adherence	100% compliance; 0 critical hosts isolated	PASS (INVIOLABLE)

Scientific Transparency Note: While Dueling DQN achieves statistically significant cost reductions over standard practice (+64.3% vs ALLOW, +28.3% vs Single-Threshold, p<0.001), it matches rather than exceeds a calibrated two-tier threshold baseline (+0.18%, p=0.7600), missing the pre-registered 15% practical target. In 4-action mode, rapid switching between RATE_LIMIT and ISOLATE also exceeds the chattering threshold (ACI=0.0214); this is eliminated in 3-action mode (ACI=0.0019).

CONCLUSION AND FUTURE WORK

- **Conclusion:** Successfully verified an end-to-end autonomous defense architecture. Random Forest delivers sub-millisecond detection (<0.80 ms, F1=0.9512). The deterministic safety gate enforces 100% critical asset uptime, while Dueling DQN delivers superior cost-optimal response selection over single-threshold systems without kernel tampering.
- **Future Work:** Hardware offload with DPDK / eBPF XDP for 10Gbps+ line-rate wire capture; cooperative multi-agent defense; and transformer-based offline cross-flow threat correlation.

APPLICATIONS

① **Enterprise SOCs:** Real-time alert triage & automated Tier-1 incident containment.
② **Cloud VPCs & SDNs:** Dynamic security group isolation & egress traffic shaping.
③ **Smart Cities & IoT:** Edge gateway anomaly filtering with zero collateral outages.
④ **Critical Infrastructure:** Inviolable allowlisting & OT boundary protection.
⑤ **Defense Networks:** Sovereign active defense with explainable Shapley audit trails.

DEMO VIDEO

PROJECT VIDEO

Scan to watch the full project demonstration video:

- Wire-speed packet capture and PCAP stream replay
- Sub-millisecond Random Forest attack detection (<0.80 ms)
- Real-time TreeSHAP waterfall feature attributions (<2.5 ms)
- 4-tier Dueling DQN response execution under safety gate
- Interactive Streamlit SOC Dashboard (Port :8501)

FEEDBACK

VISITOR SURVEY

Scan to share your feedback and comments:

- We welcome feedback and evaluations from CSE faculty, industry evaluators, and visitors at Open Day 2026.
- Venue: Academic Block A, Lab A 204 | JSSATE Bengaluru
- Project Code & Research Artifacts: github.com/rishikeshgowdakk/fyp-xrlids
- Department of Computer Science & Engineering, JSSATE Bengaluru

SATURDAY
31 OCTOBER 2026

9:30 AM – 4:30 PM

Academic Block A,
Labs and Classrooms,
JSSATE Bengaluru

Code. Create. Innovate. for a Better Future.